

SAPIENZA **ADVISOR**

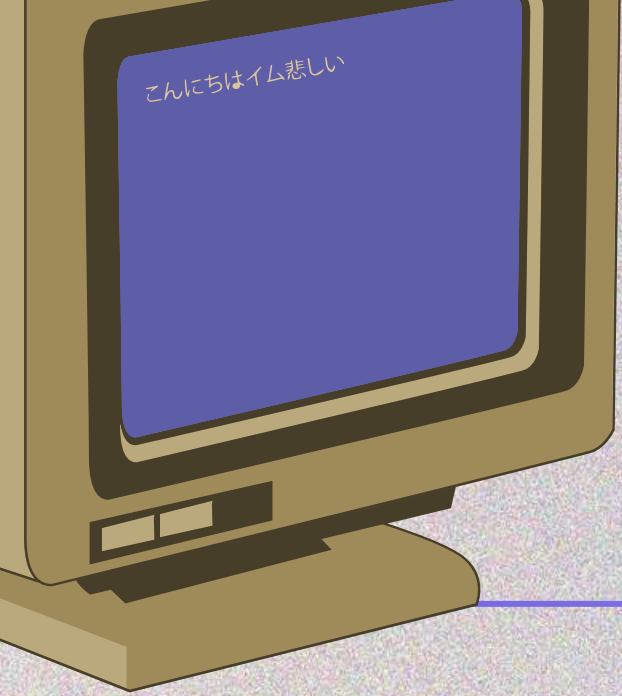
FEDERICO TURRINI

BOGDAN ANDREI TUTUIANU

LEONARDO RICCA

IGOR PASQUINI





IDEA

This project proposes an academic platform specifically for Sapienza University students that combines course evaluation with collaborative learning features.

The platform allows students to register by degree program and provides a personalized dashboard showing all their program's courses. Students can rate and review courses across multiple criteria to help peers make informed decisions. Additionally, the platform enables collaborative learning through a note-sharing system where students can upload course materials with descriptions, and these resources are peer-rated to maintain quality.

The target users include undergraduate and graduate students, academic advisors, and faculty members. The main goals are to improve course selection through peer feedback, facilitate resource sharing, and create a collaborative academic community that enhances the overall university experience.



COMPETITOR ANALYSIS

FEATURE	SAPIENZA ADVISOR	GOOGLE CLASSROOM	MOODLE	STUDOCU	PIAZZA	DISCORD / TELEGRAM	GOOGLE DRIVE
Degree Program Personalization	✓	✗	✗	✗	✗	✗	✗
Course Evaluation & Reviews	✓	✗	✗	✗	✗	✗	✗
Collaborative Note Sharing with Peer Review	✓	✗	✗	✓ (limited, no peer review)	✗	✗	✗
Integration with Academic Advisors & Faculty	✓	✓ (teacher-centric)	✓ (teacher-centric)	✗	✗	✗	✗
Structured Repository per Course	✓	✗	✓ (limited)	✗	✗	✗	✗
Community-Oriented Academic Support	✓	✗	✗	✗	✓ (Q&A only)	✓ (informal)	✗

USER PROFILES



- **Age:** 18-28 years old
- **Gender:** Diverse representation across all genders
- **Education Level:** Bachelor's to PhD students
- **Location:** Primarily Rome-based, with consideration for international exchange students
- **Technology Proficiency:** High smartphone and computer literacy, native digital experience
- **Academic Status:** Active enrollment in university programs
- **Motivation:** Seeking efficient study methods, peer collaboration, and informed academic decision-making

PERSONAS **and** **SCENARIOS**

SOFIA ROMANO, 21

BIOMEDICAL ENGINEERING STUDENT



She lives with her parents and younger brother in Monteverde, a quiet neighborhood she's very fond of. Despite studying Biomedical Engineering at Sapienza, she keeps her academic life well balanced with her personal interests and routines.

Sofia is a calm and introverted person, but very empathetic and attentive to the people around her. She doesn't like crowded places and prefers intimate settings—afternoons at a café, a walk with a friend, or a quiet evening watching films.

She has a passion for photography and always carries her analog camera when she explores the city. She enjoys discovering hidden spots in Rome, especially early in the morning when the streets are still quiet.

Her closest friends describe her as reliable and grounded. She's the one who remembers birthdays, offers thoughtful advice, and always checks in after a tough day. Even though she has lived in Rome her whole life, she dreams of doing an Erasmus to experience something new—without losing touch with her roots.

SCENARIO



Three weeks before her Biomechanics exam, after her morning swim and a quick breakfast with her family, Sofia struggles with complex mathematical proofs and theoretical concepts. Previous attempts to find study materials through social media groups yielded fragmented, unreliable information, eating into the precious time she had allocated for actual studying before her planned weekend trip to Bologna.

Accessing SapienzaAdvisor, Sofia filters available materials by course, rating, and content type. She quickly identifies highly-rated summary notes that received positive peer reviews for clarity and comprehensiveness. The platform's quality assurance system, based on student ratings and academic verification, ensures she accesses reliable content without wasting hours verifying accuracy. Additionally, Sofia finds supplementary exercise sets with detailed solutions, uploaded by students who successfully completed the course in previous years. The platform's review system helps her identify the most effective study strategies and connect with peer study groups preparing for the same examination. She schedules a group study session at the engineering library, knowing she can maximize her limited study time with quality resources and peer support.

DAVID MÜLLER, 23

EXCHANGE STUDENT FROM GERMANY



He comes from a town near Munich and is currently in Rome for his Erasmus at Sapienza University, where he studies Modern Languages and Linguistics, focusing on Italian, Spanish, and intercultural communication. David is thoughtful and gentle, with a natural curiosity for cultures and human connections. He's a good listener, and although a bit introverted at first, he quickly warms up in one-on-one conversations.

He shares an apartment in Piazza Bologna with other Erasmus students, and enjoys the mix of languages and lifestyles. In his free time, he likes reading novels in different languages, writing short travel notes, and taking long walks through Rome's historic neighborhoods with his analog camera.

David is fascinated by everyday life in Italy—small cafés, local expressions, the rhythm of conversation. For him, this Erasmus is a chance not only to improve his language skills, but to live fully immersed in a different way of seeing the world.

SCENARIO



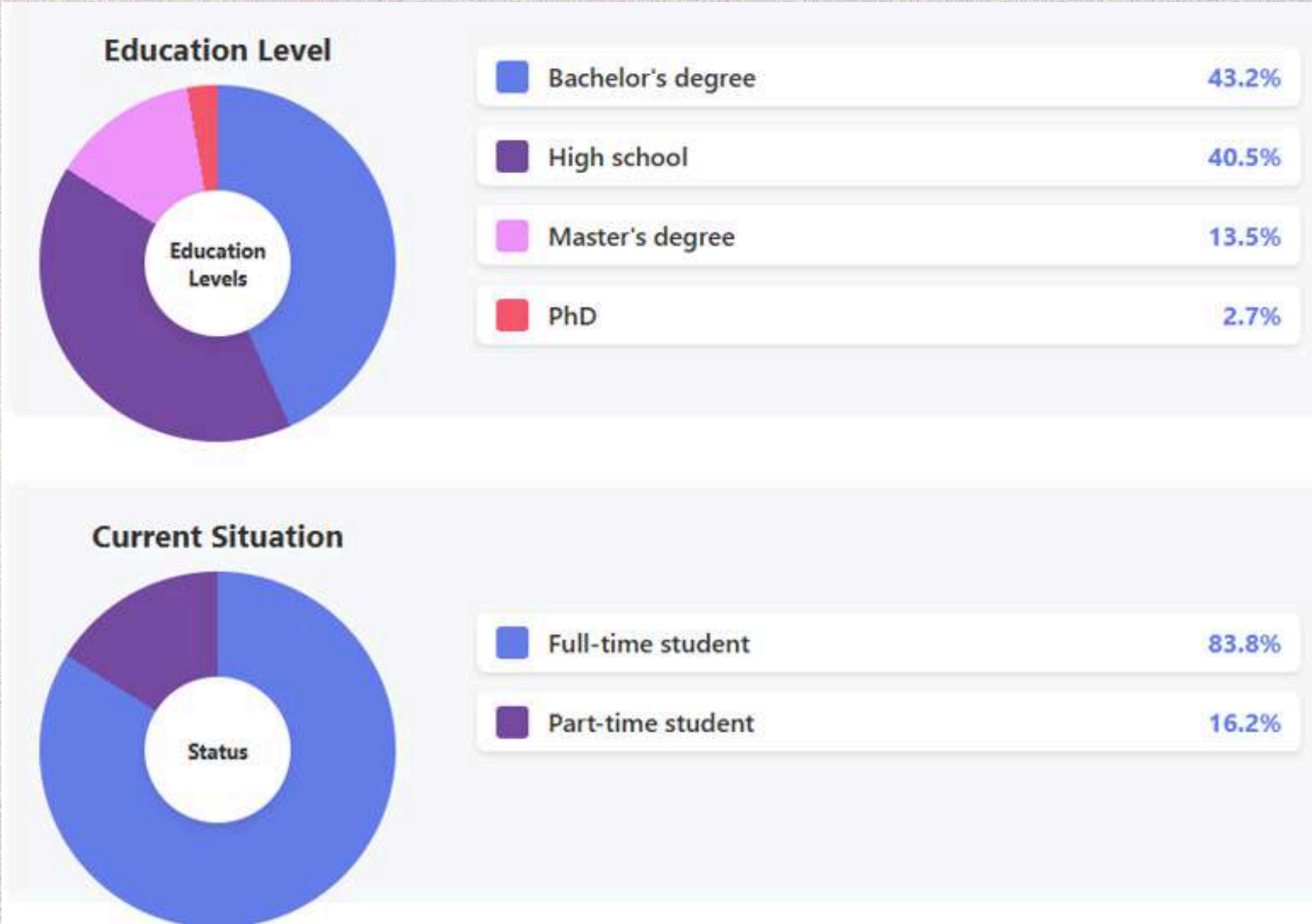
Two weeks into his exchange program, after a morning spent photographing the Colosseum at sunrise, David returns to his residence needing to finalize his course selection while ensuring compatibility with his home university requirements. Language barriers make it difficult to fully understand course descriptions and academic expectations, especially the oral examination format that's uncommon in Germany. Using SapienzaAdvisor's international student features from his laptop in the residence's common room, David filters courses by language of instruction and accesses English-language reviews and materials. The platform's detailed course descriptions, enhanced by peer reviews from previous international students, provide clarity about expectations, assessment methods, and workload. He's particularly relieved to find reviews from other German exchange students who've navigated the same credit transfer challenges.

Through the platform, David connects with other international students and finds study materials translated or created in English. He discovers that one highly-rated economics professor conducts consultations in English and is known for being supportive of exchange students. The community aspect of SapienzaAdvisor helps him integrate into the academic environment while maintaining academic success aligned with his home university requirements.

QUESTIONNAIRE ANALYSIS

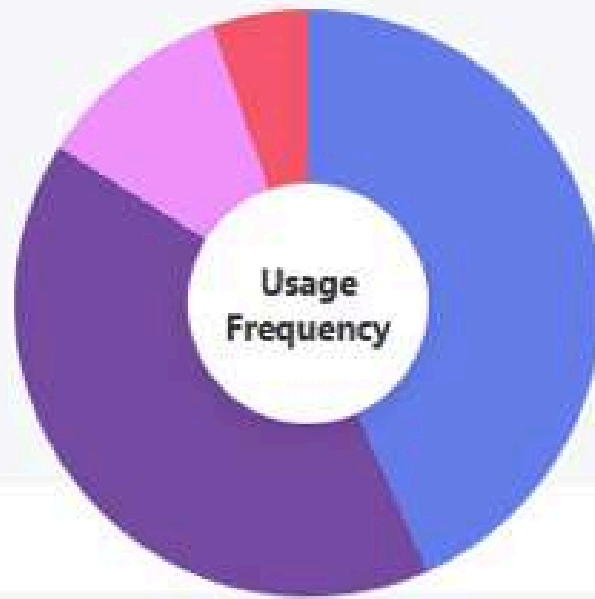
Our comprehensive questionnaire received **37** responses from students across various university programs.

DEMOGRAPHICS OVERVIEW



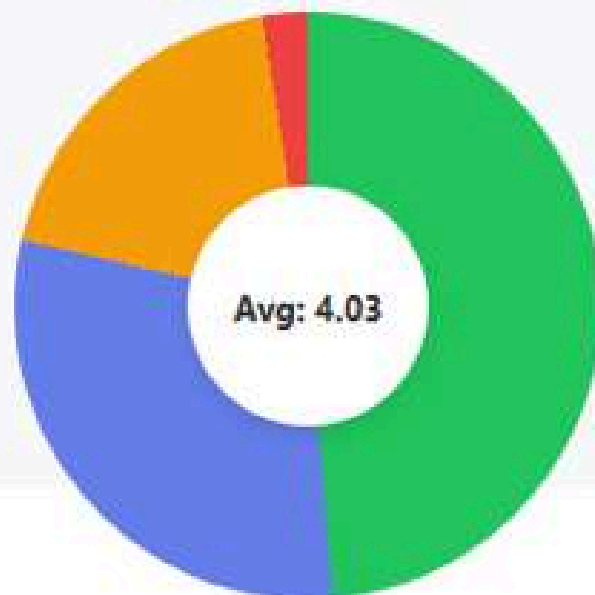
DIGITAL HABITS AND COMFORT

Online Resources Usage Frequency



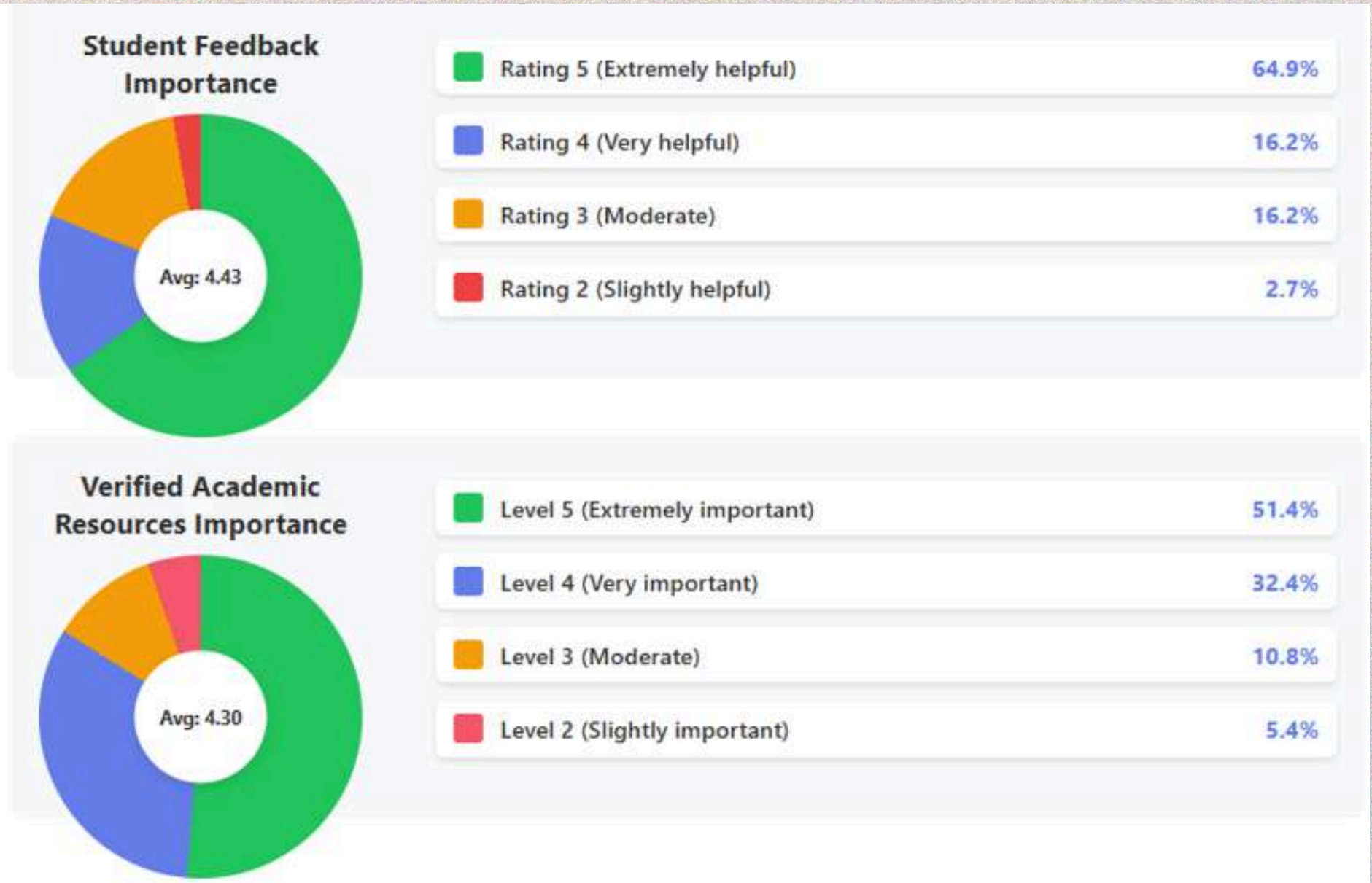
Always	43.2%
Often	40.5%
Rarely	10.8%
Sometimes	5.4%

Digital Platform Comfort Level



Level 4 (Comfortable)	48.6%
Level 5 (Very comfortable)	29.7%
Level 3 (Moderate)	18.9%
Level 1 (Not comfortable)	2.7%

PLATFORM VALUE ASSESSMENT



SOCIAL BEHAVIOUR AND TRUST



COURSE SELECTION CRITERIA

What students consider most important when choosing courses (multiple selections allowed)



CHALLENGES FACED CHOOSING COURSES

Primary challenges students face when choosing university courses

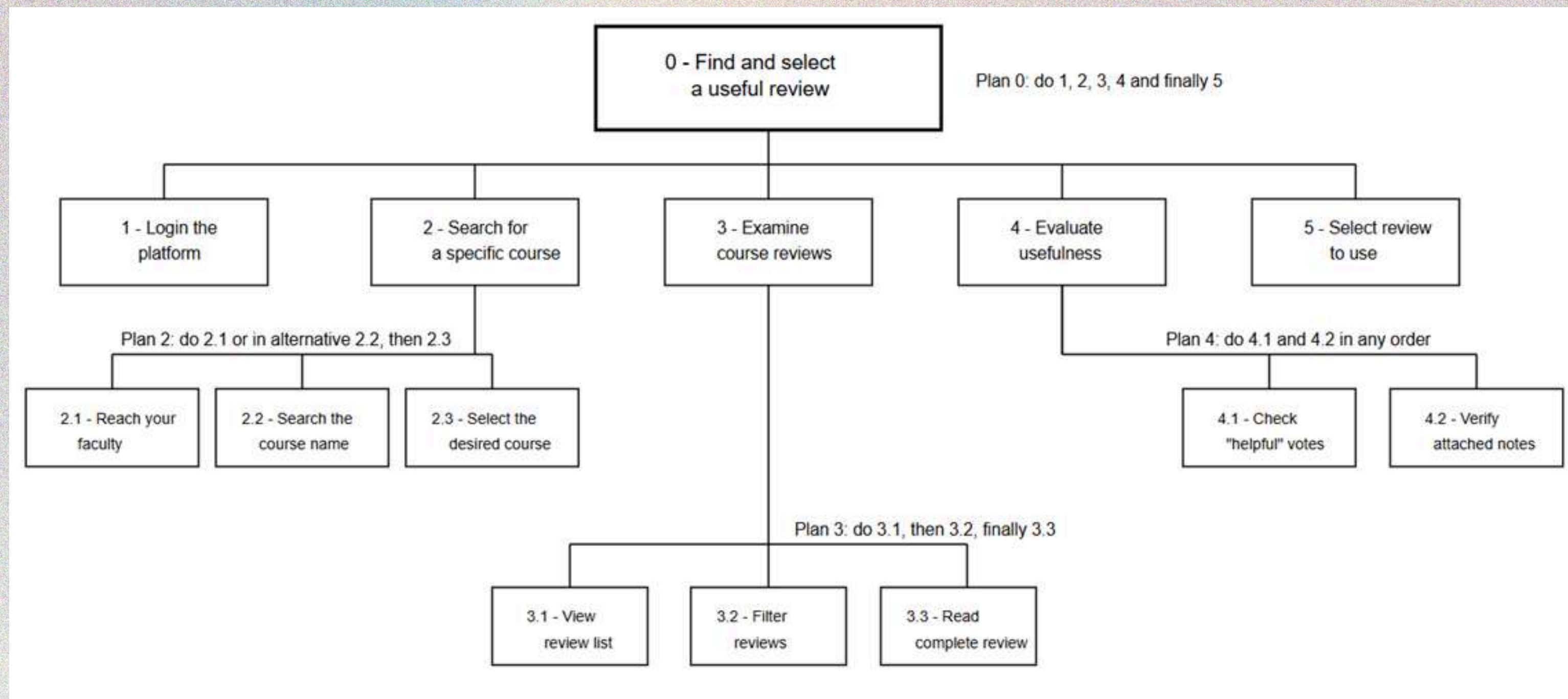


TASK ANALYSIS

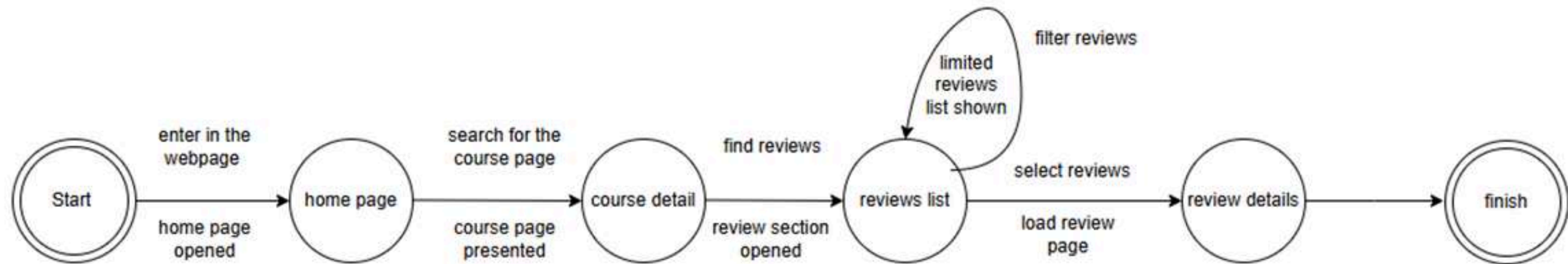
HTA & STN



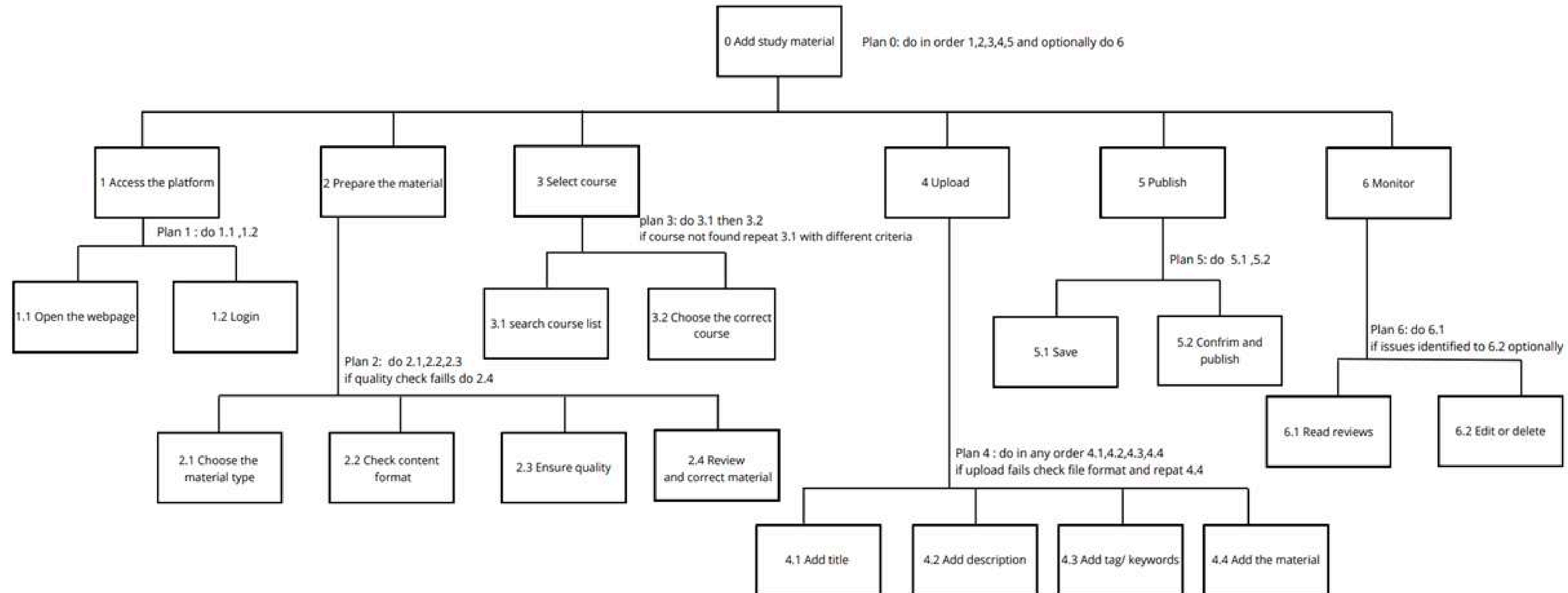
HTA FOR “FIND AND SELECT A USEFUL REVIEW FOR A UNIVERSITY COURSE”



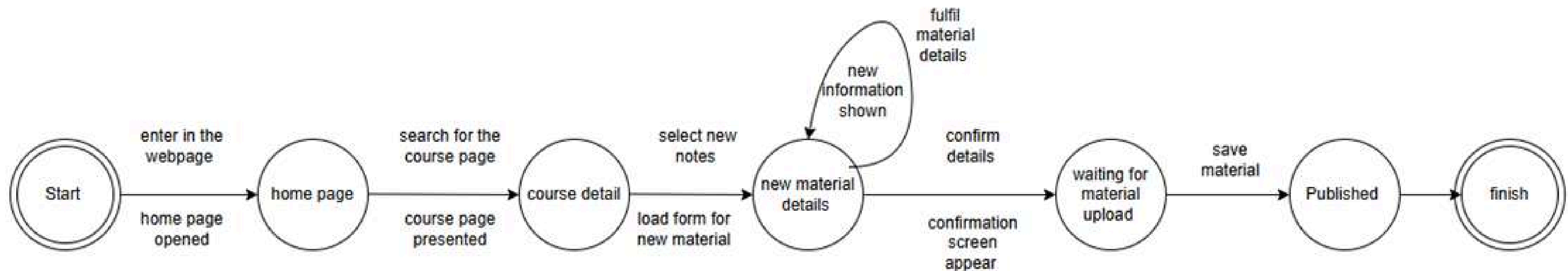
STN FOR “FIND AND SELECT A USEFUL REVIEW FOR A UNIVERSITY COURSE”



HTA FOR “ADD STUDY MATERIAL TO THE PLATFORM”



STN FOR “ADD STUDY MATERIAL TO THE PLATFORM”



MOCK-UPS

REGISTRATION AND ONBOARDING FLOW

Login Page

SapienzaAdvisor

https://sapienzaadvisor.it

SapienzaAdvisor

Login

Email

Password

Forgot the password?

New account? Register

Login

Or login with

G

f

Register Page

SapienzaAdvisor

https://sapienzaadvisor.it

SapienzaAdvisor

Register

Email

Password

Name

Last Name

Birthdate

City

Have an account? Login

Register

Registration Succeeded Login Page

SapienzaAdvisor

https://sapienzaadvisor.it

SapienzaAdvisor

REGISTRATION SUCCESSFUL!

Login

Email

Password

Login

Choose your Faculty

SapienzaAdvisor

https://sapienzaadvisor.it

SapienzaAdvisor

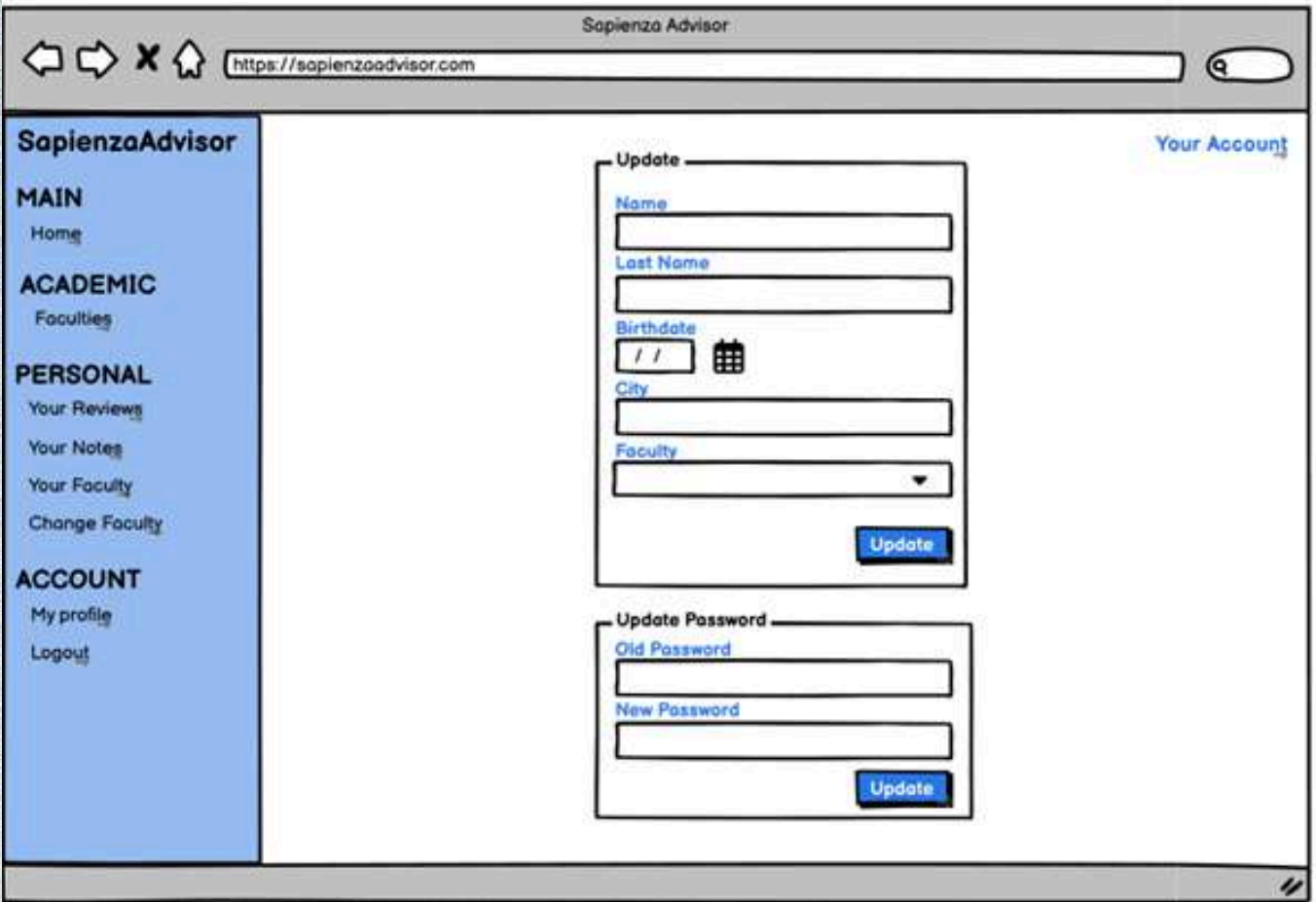
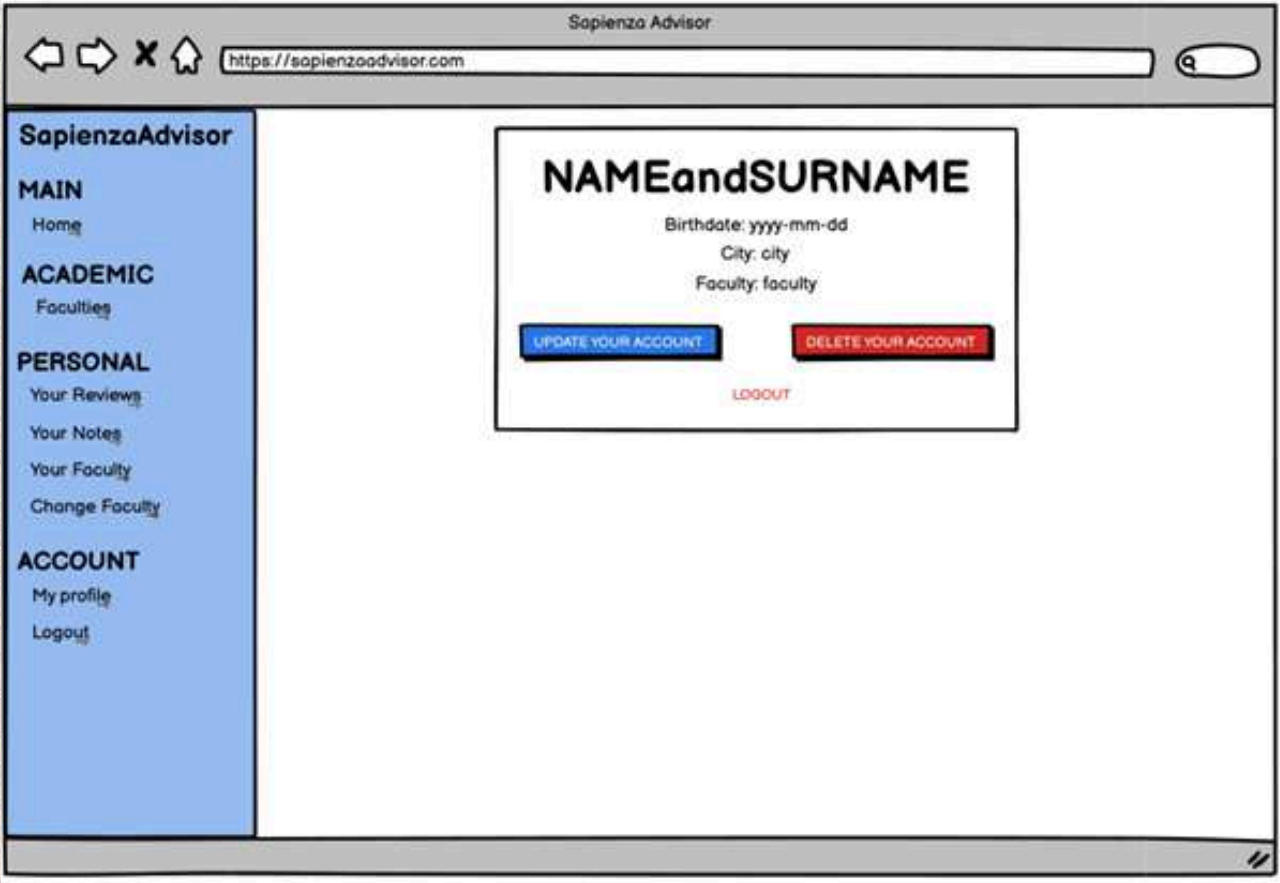
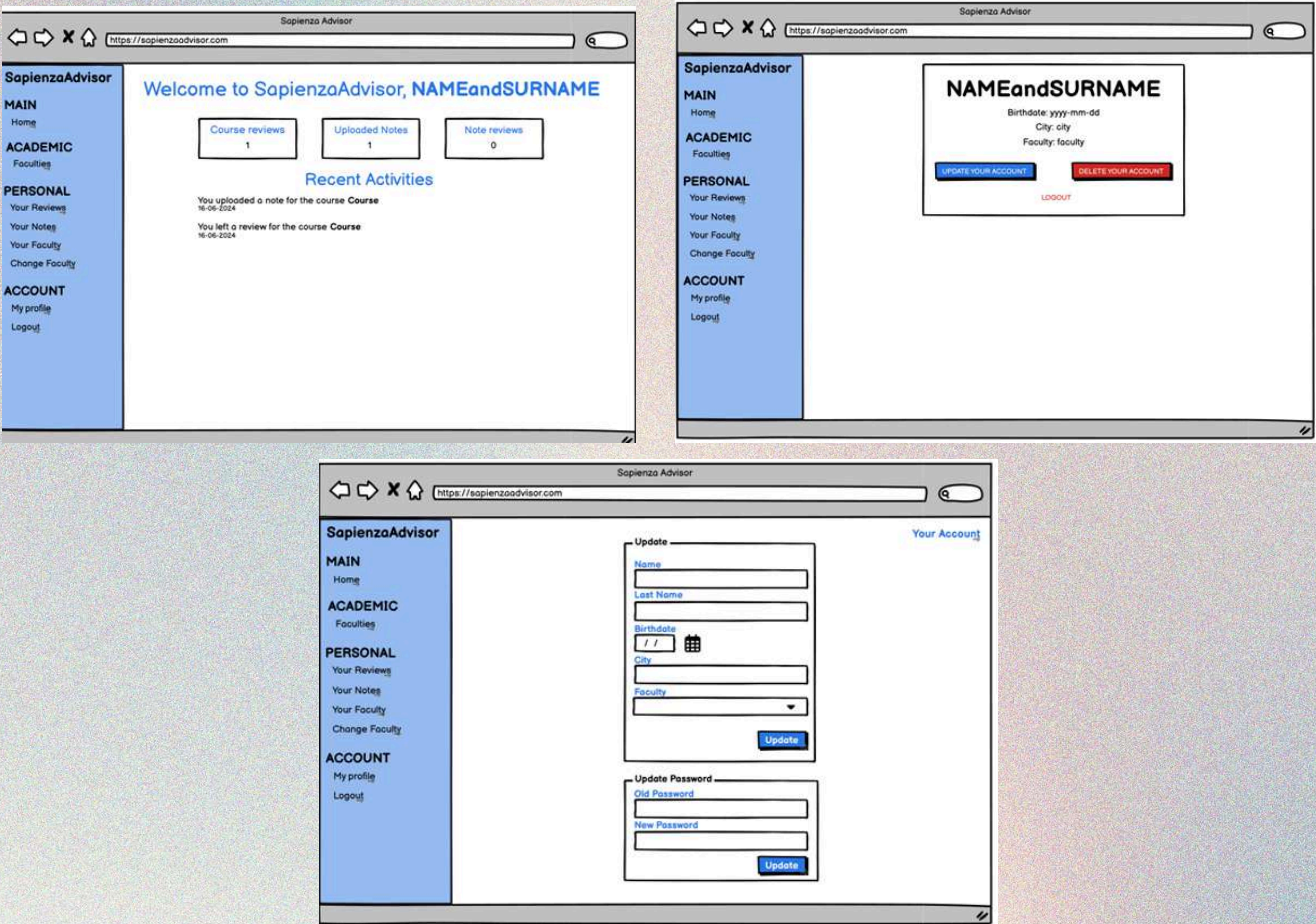
Select your faculty

Thanks for registering to Sapienza Advisor.
Please select the faculty you want to enroll in.

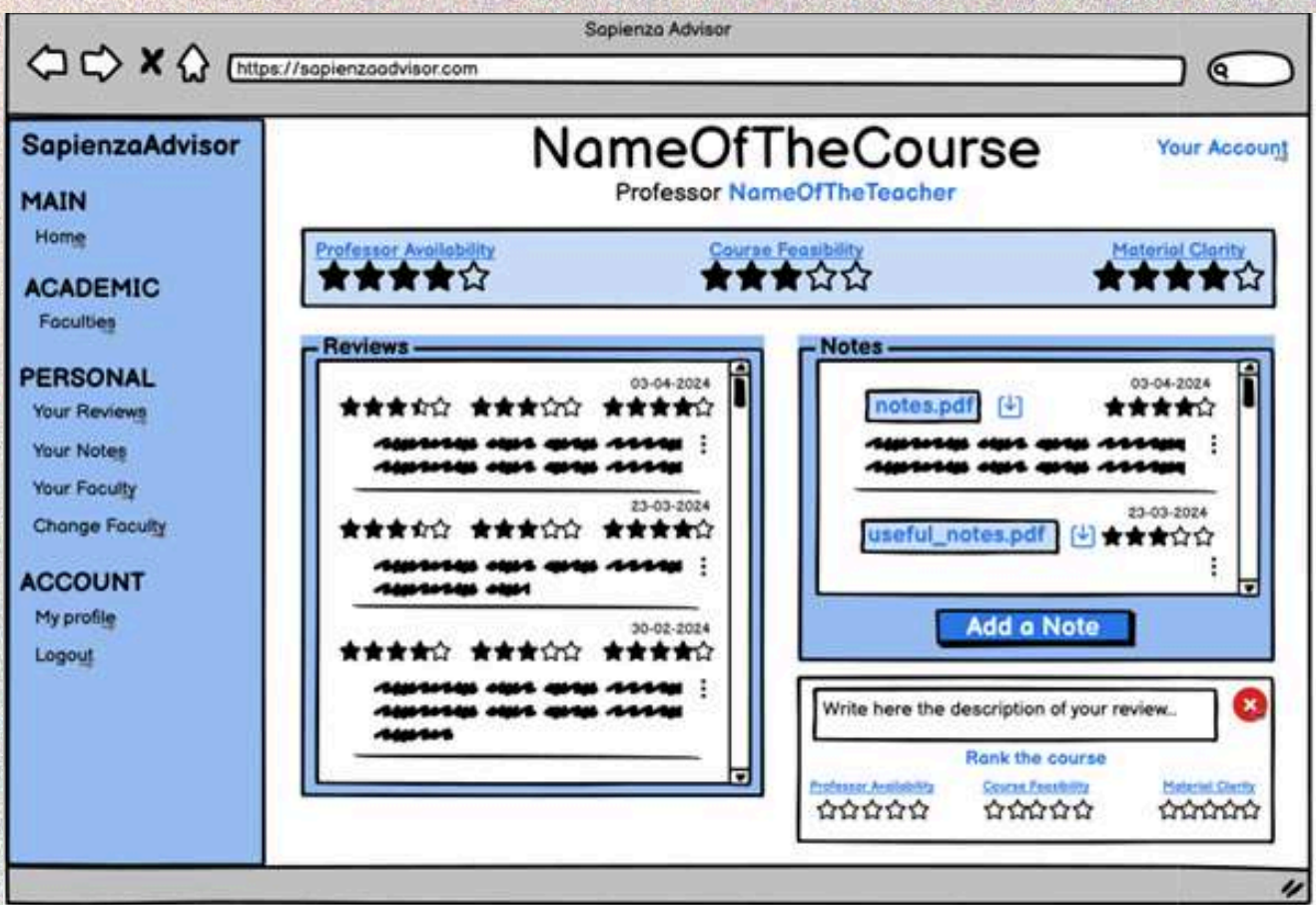
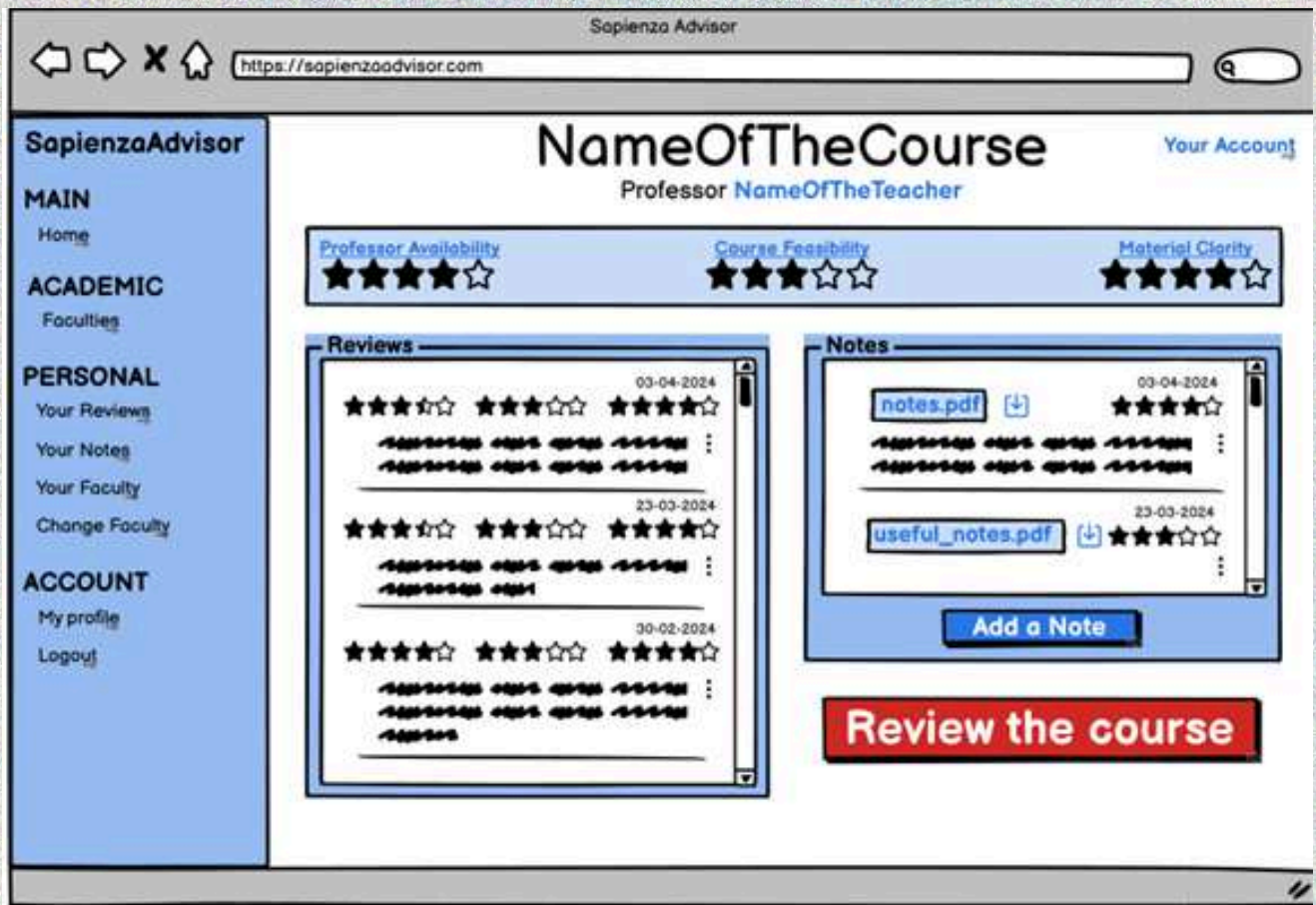
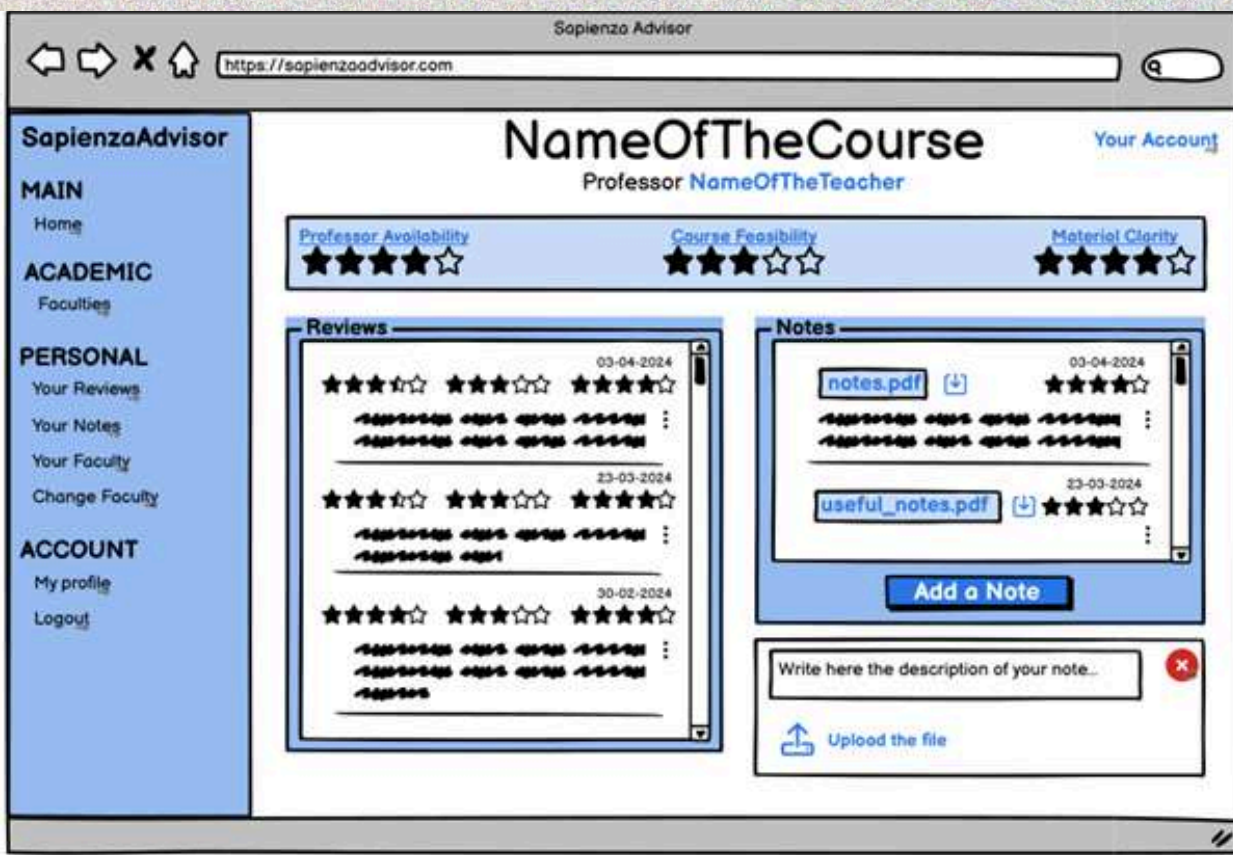
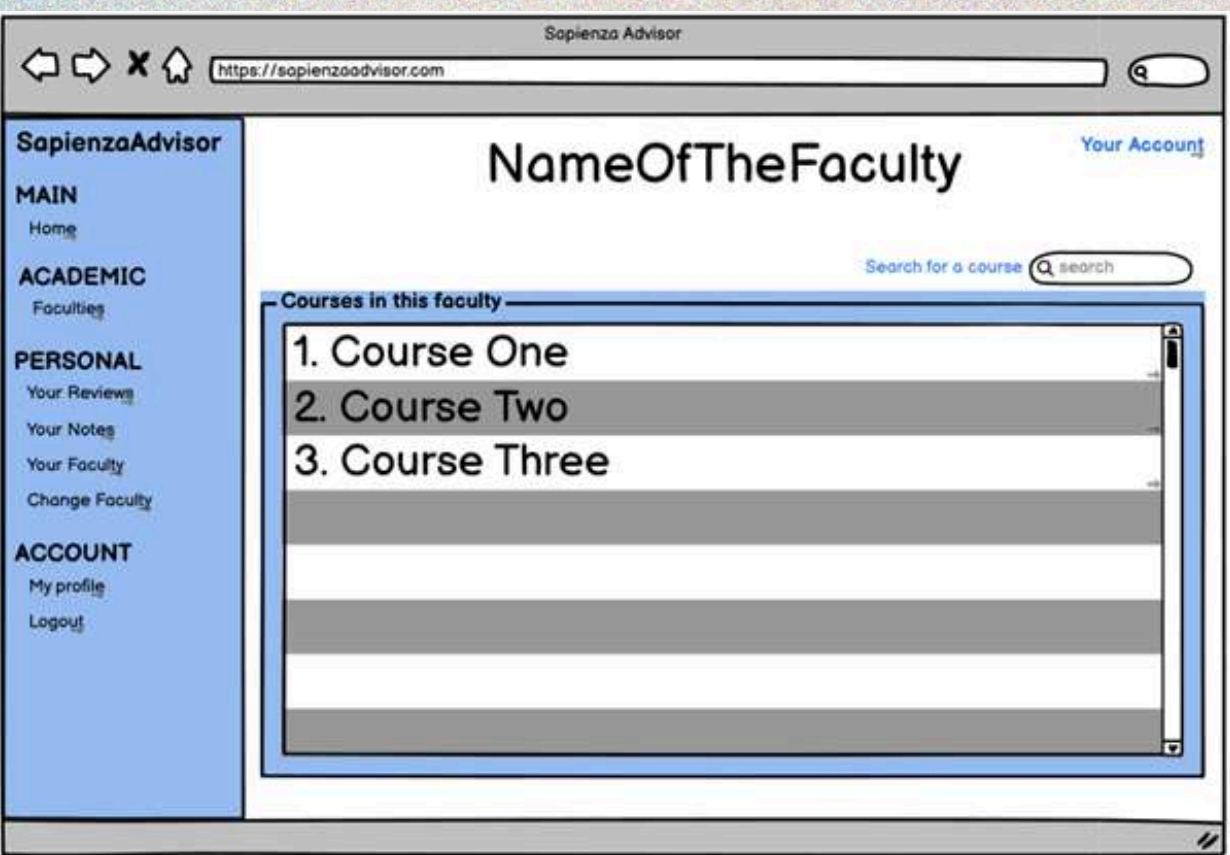
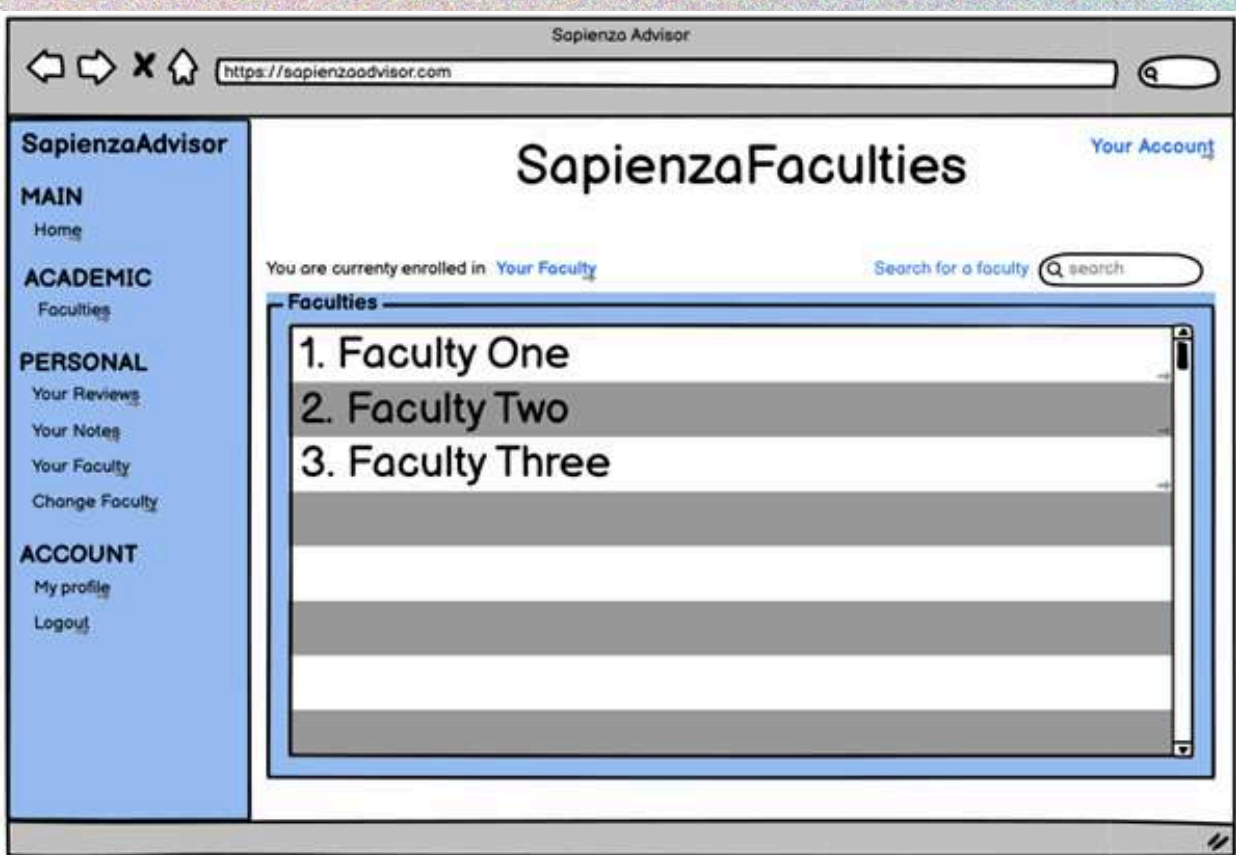
Faculty

Confirm Faculty

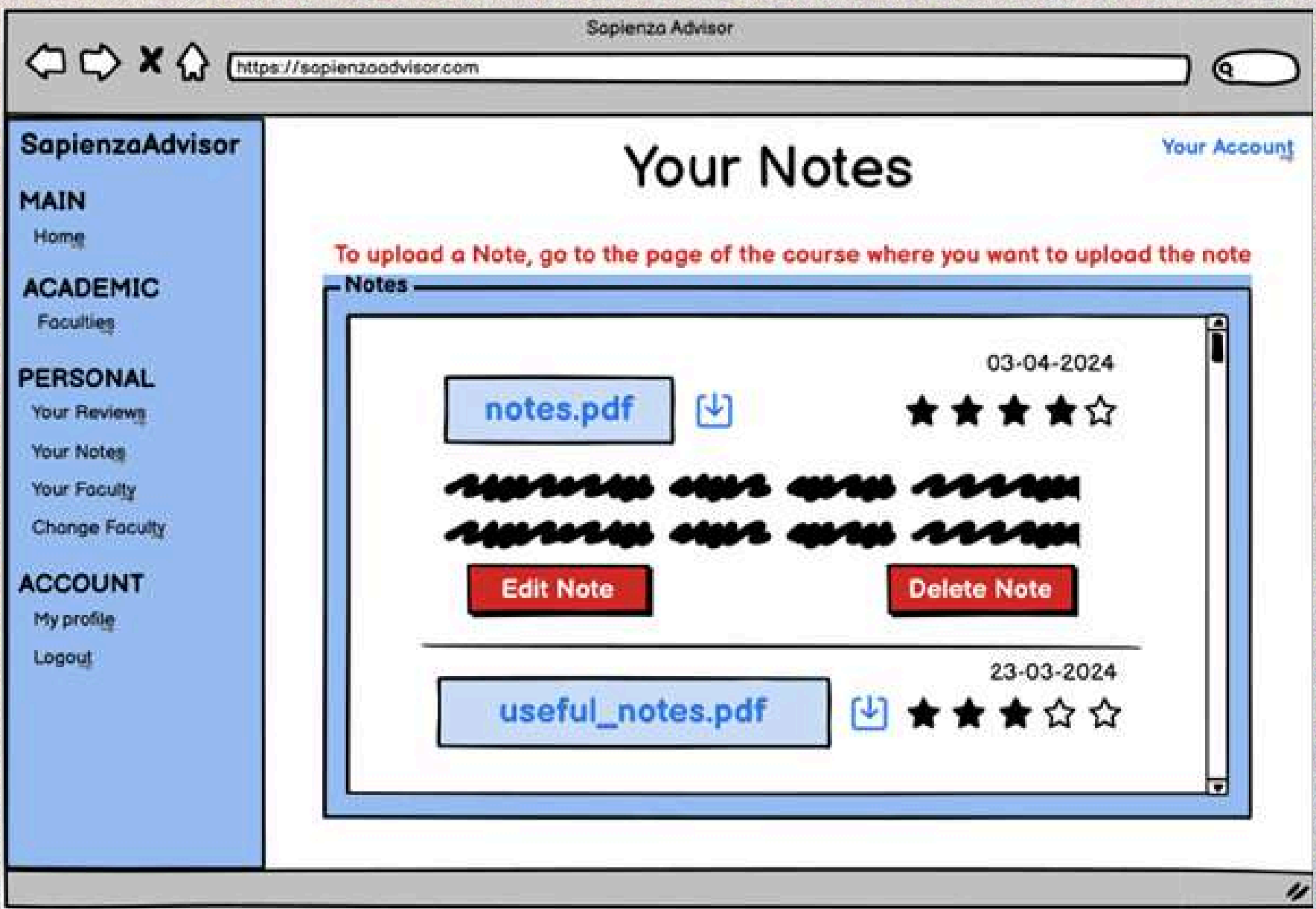
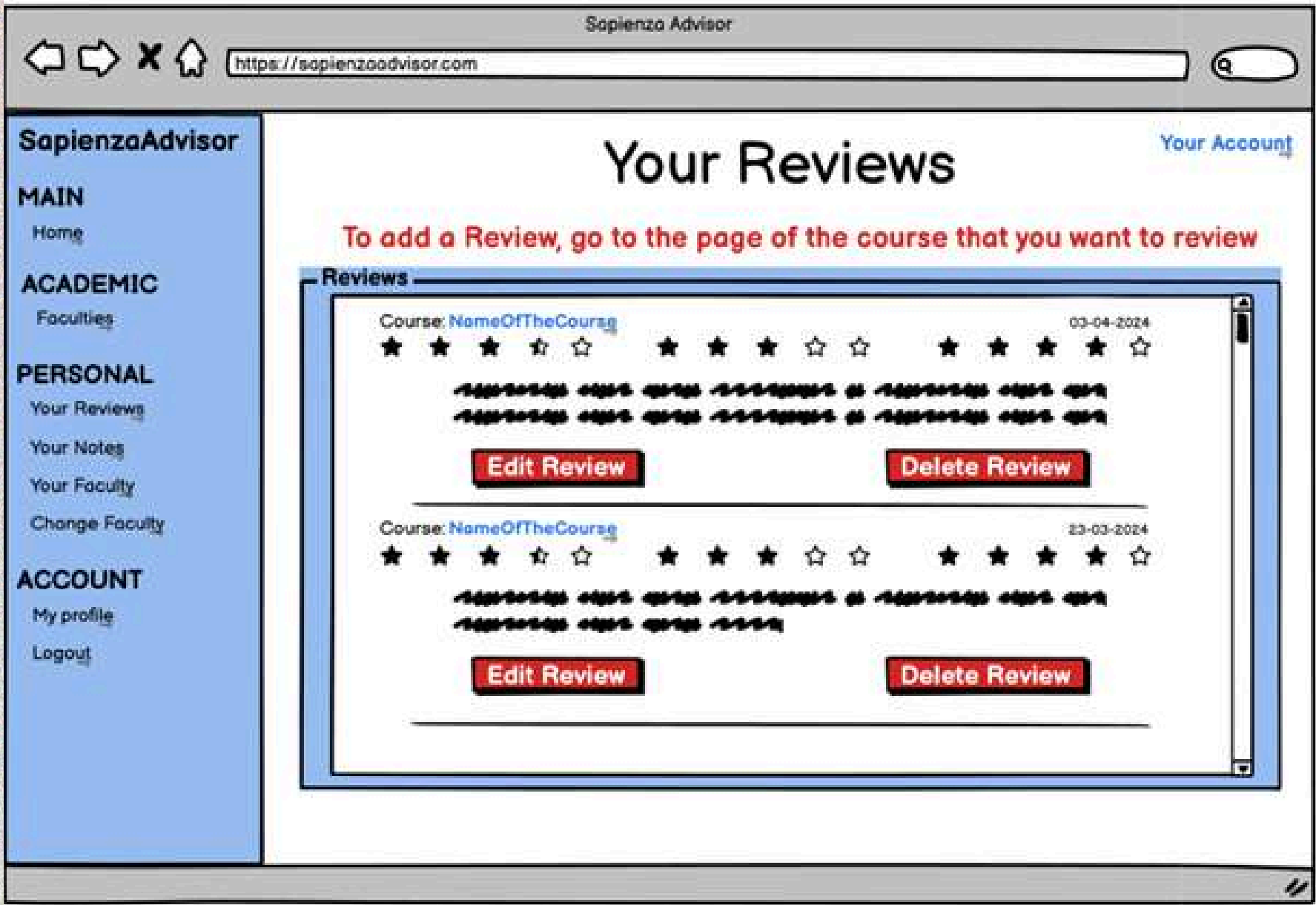
MAIN PLATFORM INTERFACE



COURSE DISCOVERY AND INTERACTION



PERSONAL CONTENT MANAGEMENT



EXPERT-BASED EVALUATION

HEURISTIC EVALUATION

Frame	Heuristic violated	Severity	Description / Comment
Login/Signup	Error Prevention	3	There is no visible validation for the email format or the requirements for the password.
Login/Register	Consistency & Standards	2	The "Forgot Password," "New Account? Register," and "Have an account? Login" do not appear as links.
Login	Consistency and Standards	2	The social login buttons for Google and Facebook lack visual consistency.
Login/Signup	Help & Documentation	3	There is no help, onboarding, or tooltips available for new users to explain the functionalities of the site.
Update profile	Error Prevention	3	Password update lacks current password verification
Update profile	User control and freedom	2	No "Cancel" option to abandon changes
Add a note	Error Prevention	2	There are no indications regarding file format validation or size limits when adding a note.
Add a note	User control and freedom	2	There is no "Cancel" option available to close the modal without saving changes.

The evaluation of our system was made by using Jakob Nielsen's 10 heuristics. In our initial Mock-Up, we have violated a total of eight different heuristics.

Heuristic used

1. Visibility of system status
2. Match between the system and the real world
3. User control and freedom
4. Consistency and standards
5. Error prevention
6. Recognition rather than recall
7. Flexibility and efficiency of use
8. Aesthetic and minimalist design
9. Help users recognize, diagnose and recover from errors
10. Help and documentation

o = I don't agree that this is a usability problem at all

1 = Cosmetic problem only

2 = Minor usability problem

3 = Major usability problem

4 = Usability catastrophe

ERROR CORRECTION AND FIRST PROTOTYPE

Missing email format validation and password requirement indicators (Severity 3)

Register

Email

Password

Name

Last Name

Birthdate

City

Have an account? [Login](#) [Register](#)

Email

wrongemail.,@gmail.com

Invalid email format

Password

test33

- At least 6 characters
- Contains at least one letter
- Contains at least one number

Confirm Password

test33

Birthdate

gg/mm/aaaa

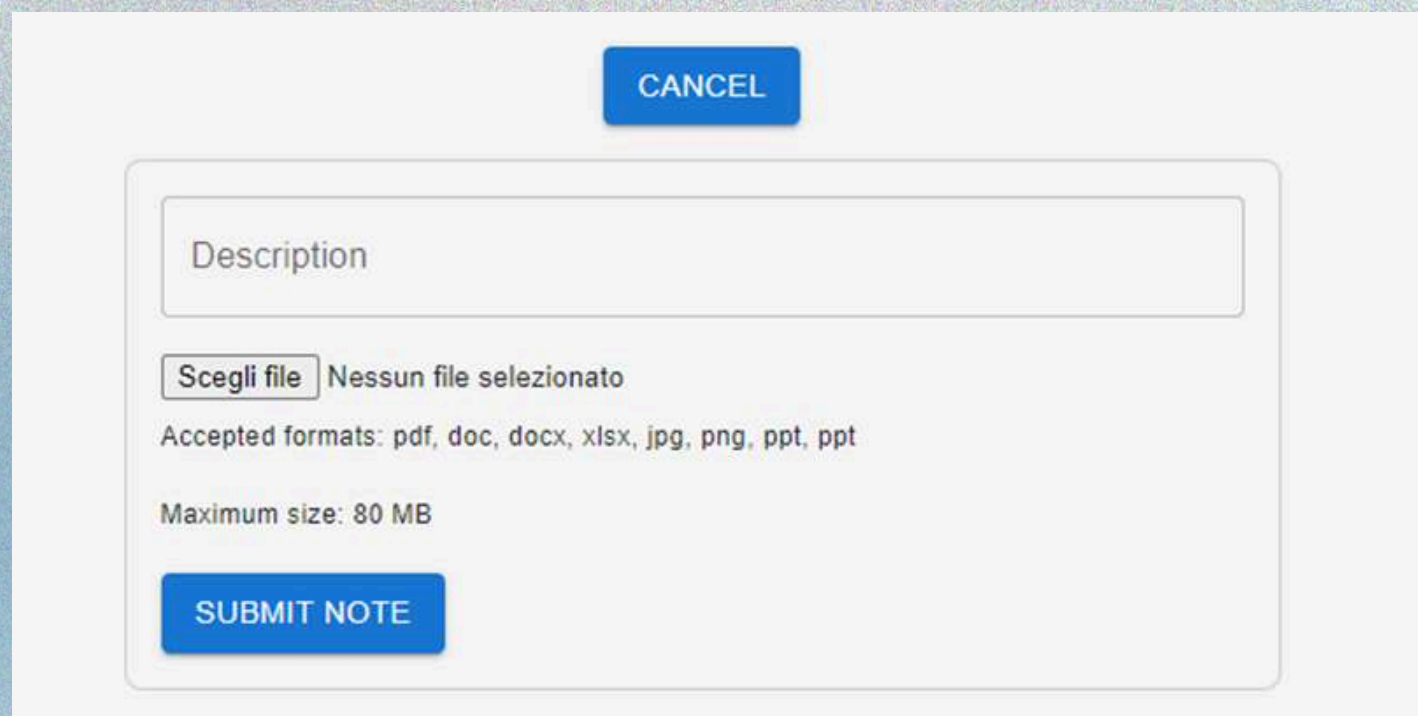
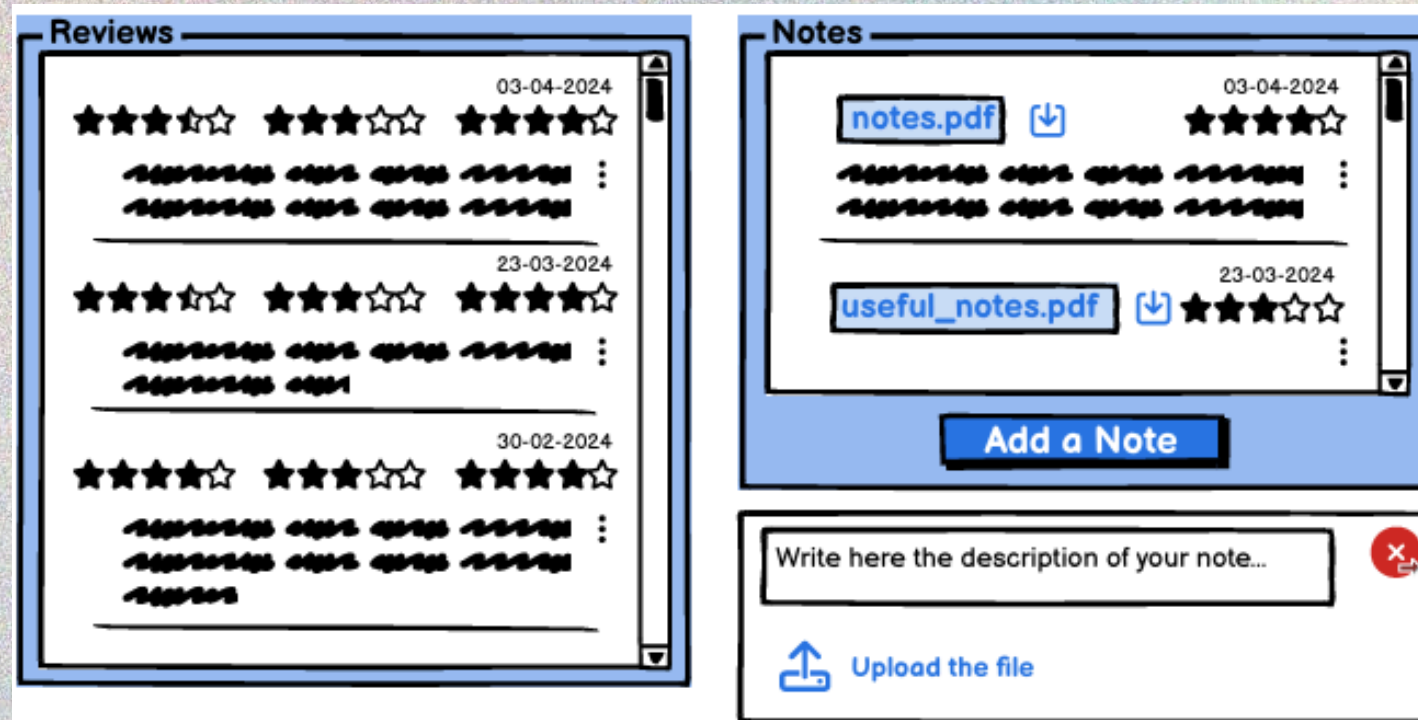
City

Register

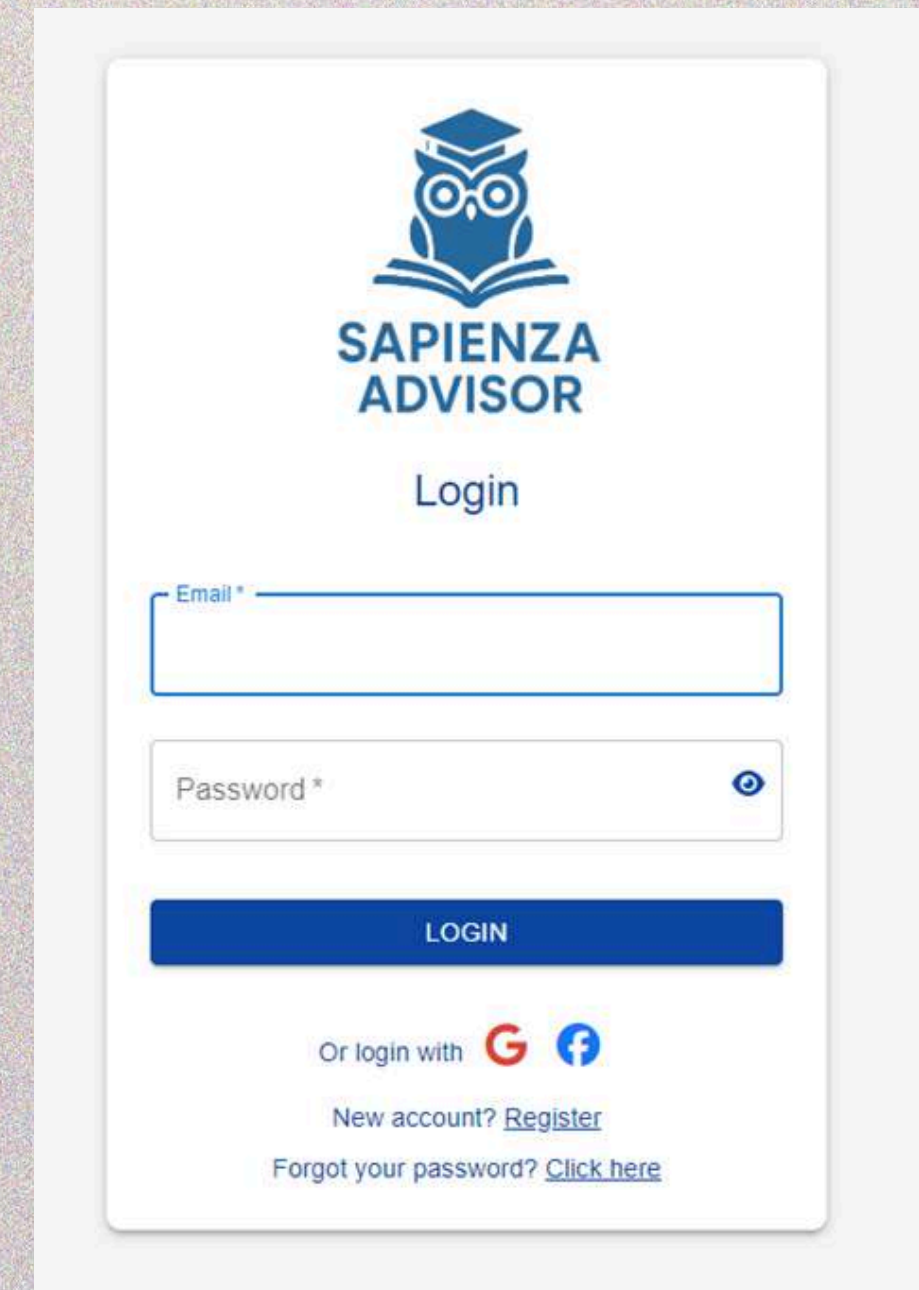
Have an account? [Login](#)

ERROR CORRECTION AND FIRST PROTOTYPE

**Missing file format and size limitation guidance
(Severity 3)**

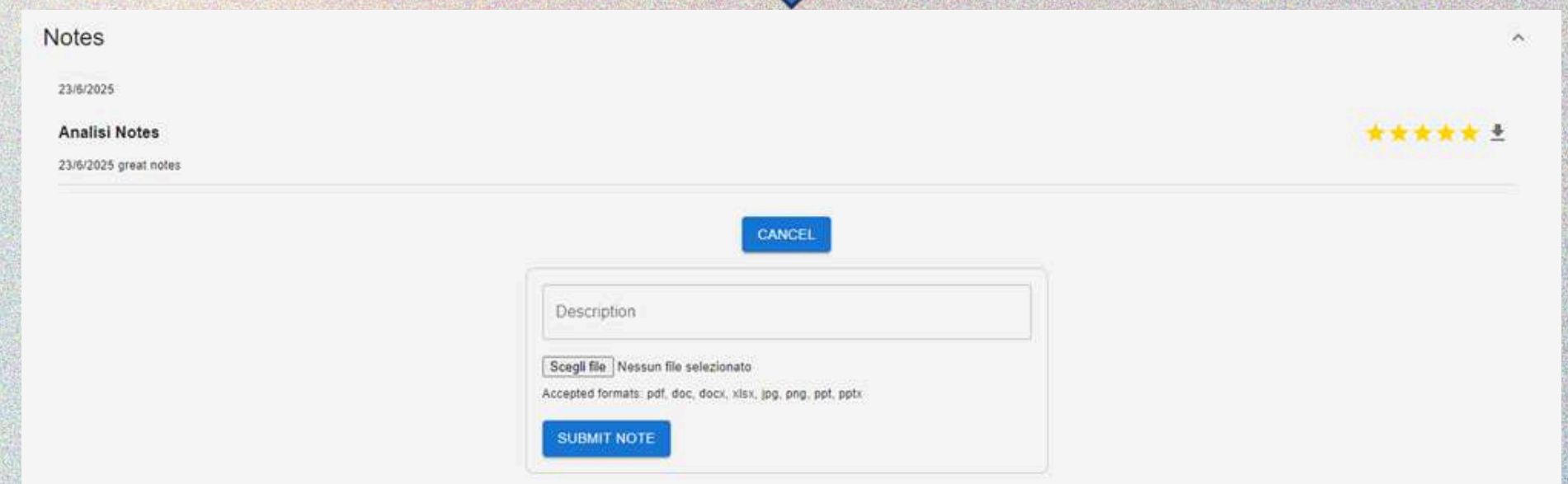
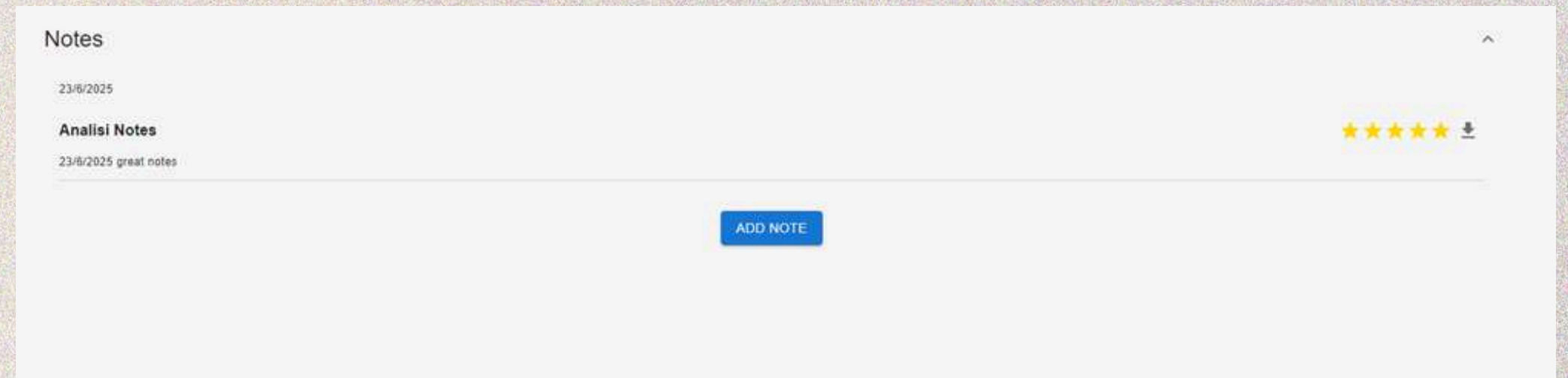
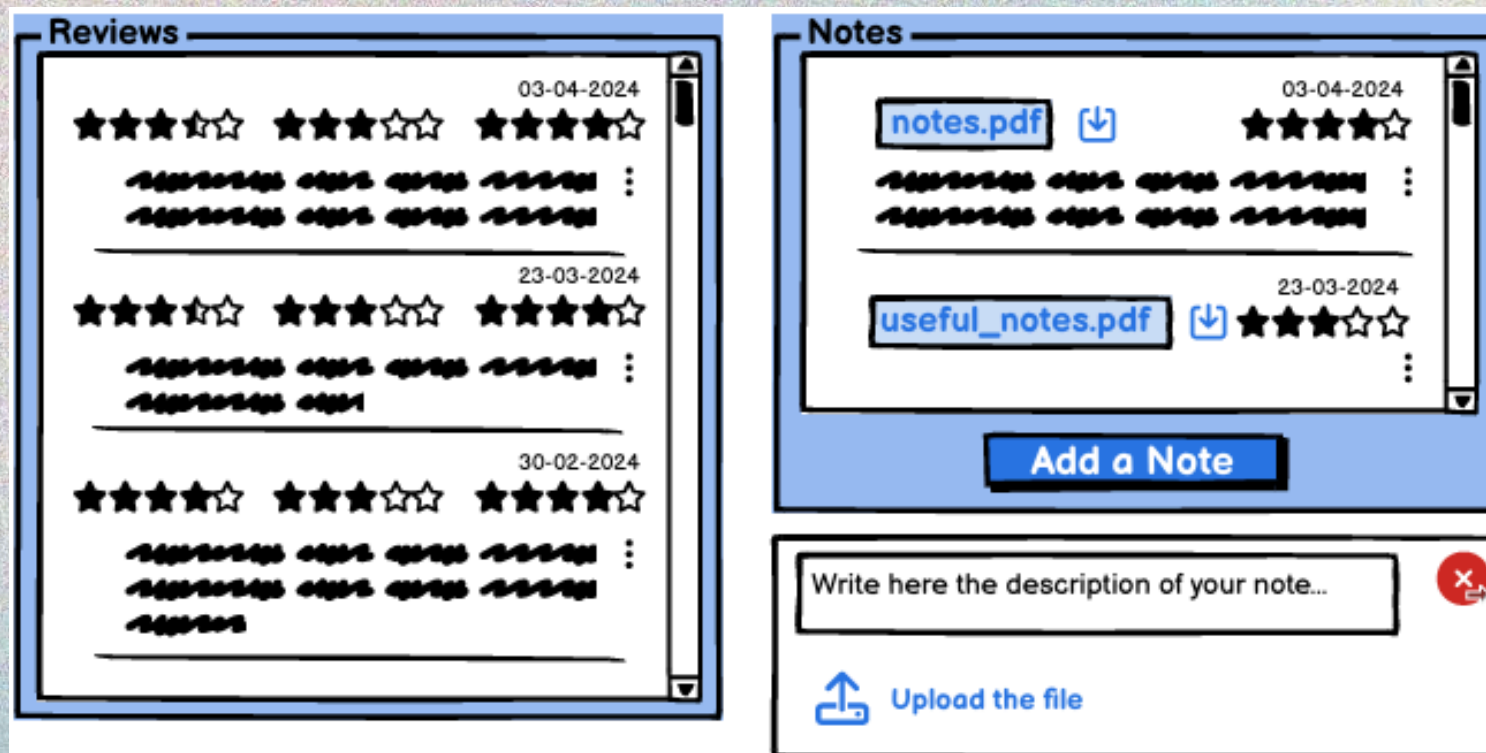


Inconsistent styling of social login buttons (Severity 2)



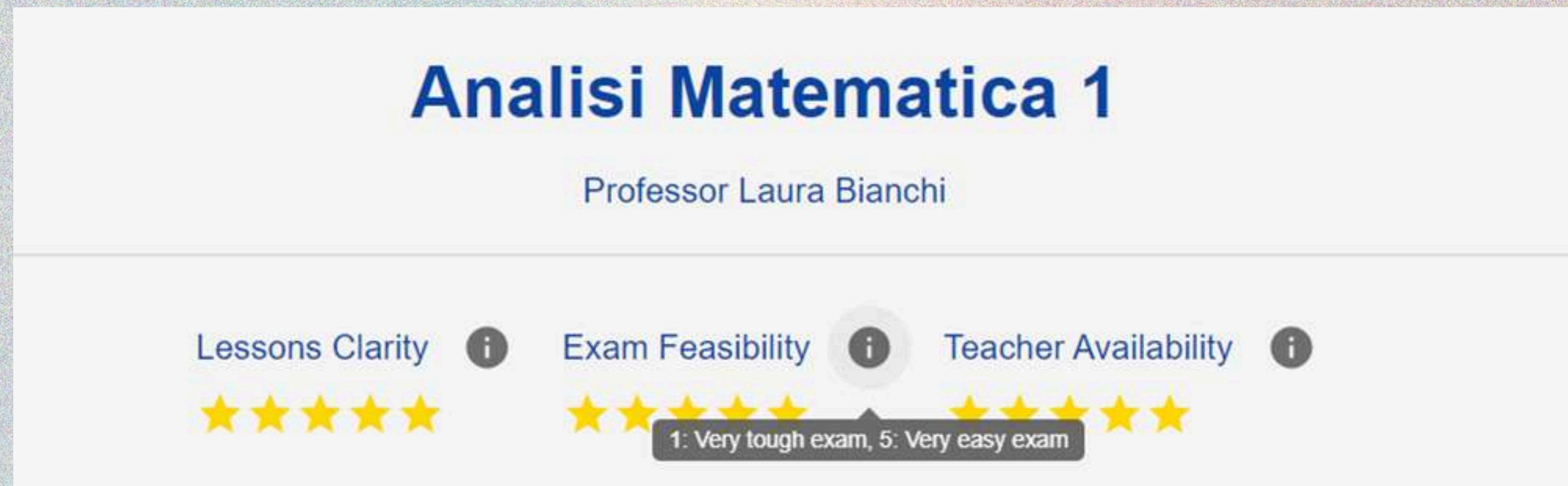
ERROR CORRECTION AND FIRST PROTOTYPE

Missing cancel options in the Notes section (Severity 2)

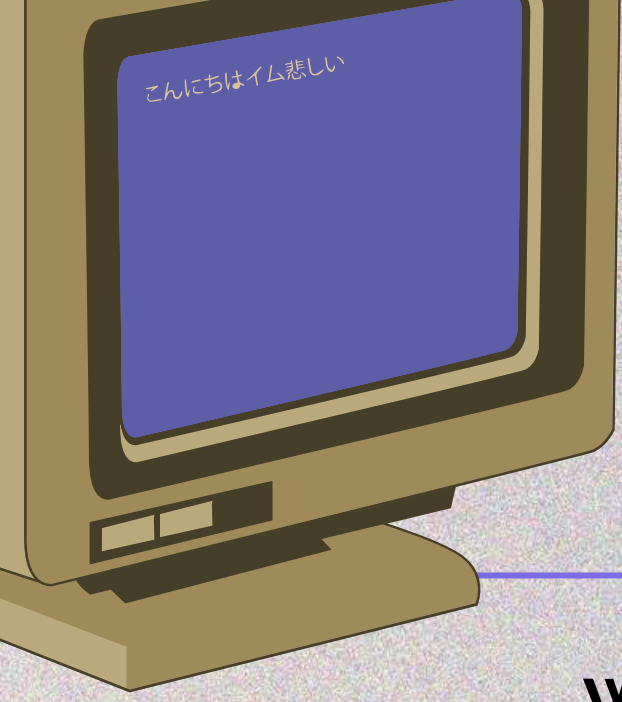


ERROR CORRECTION AND FIRST PROTOTYPE

Lack of help or tooltips to explain functionalities for new users



USER-BASED EVALUATION



THINK ALOUD PROTOCOL

We recruited 10 participants representing our target user demographics, ensuring coverage across the two primary personas identified in our requirement analysis.

Task 1: Course Review Discovery *Scenario:* "You are planning your next semester courses and need to evaluate 'Advanced Machine Learning' (or relevant course for your field) to determine if it fits your academic goals and schedule constraints."

Task 2: Study Material Upload *Scenario:* "You have comprehensive notes from a recently completed course that performed well for your exam preparation. Share these materials to help future students."



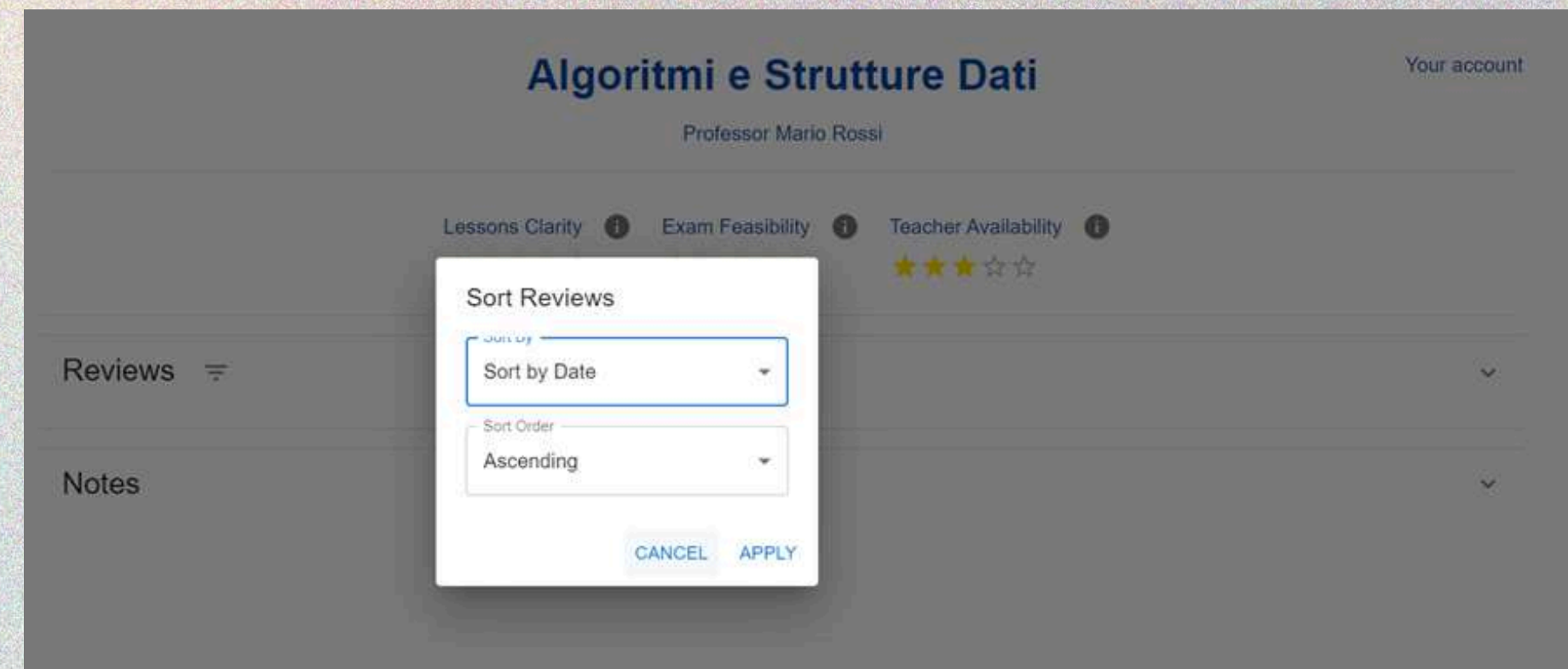
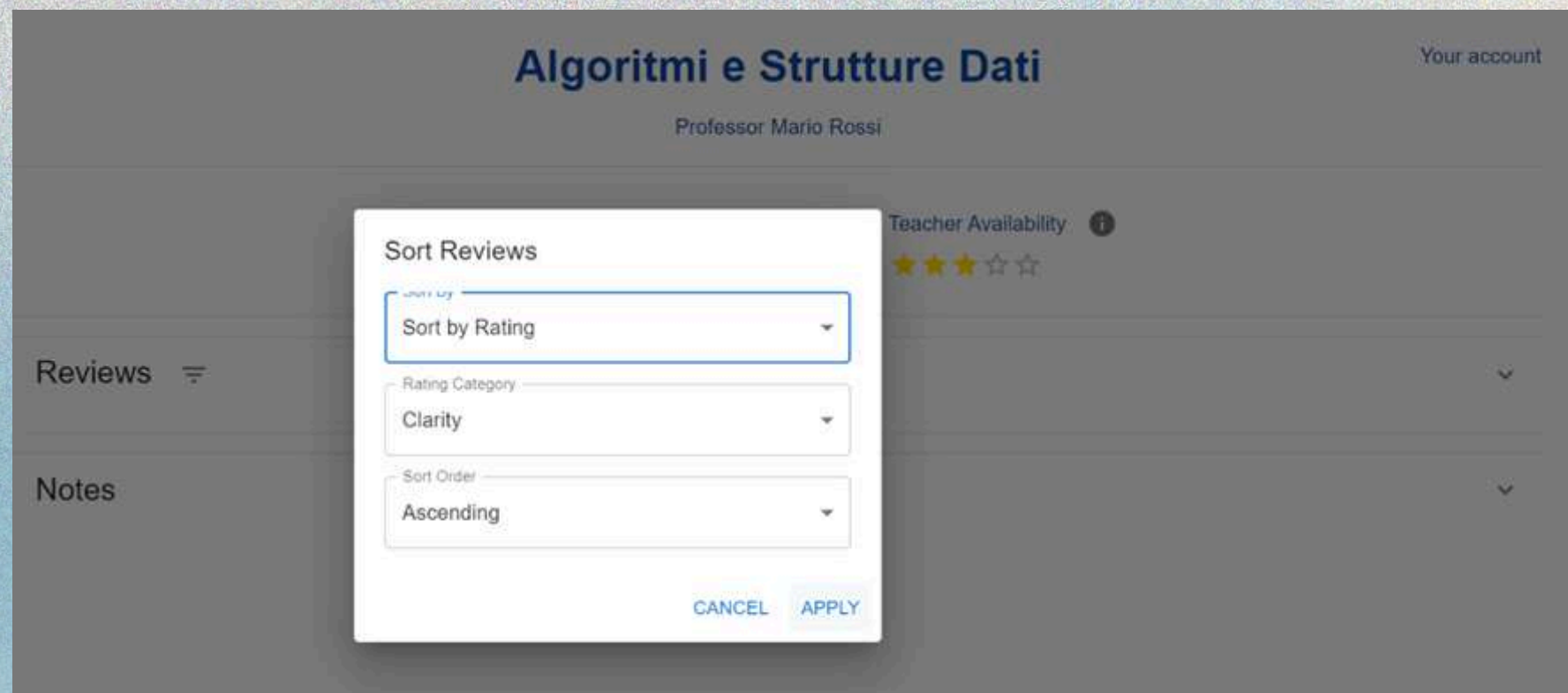
COURSE REVIEW DISCOVERY

Challenge: Overwhelming Review Volume

- Users faced difficulty navigating courses with extensive review histories
- **Feedback included:**
 - Feeling overwhelmed by detailed reviews
 - One participant felt lost and noted the need for a more systematic approach

Solution: Advanced Filtering Features

- **Implemented multi-parameter filtering:**
 - Lesson Clarity
 - Course Feasibility
 - Material Clarity
- **Enabled:**
 - Ascending/Descending sorting
 - Temporal filtering (by time/date)



STUDY MATERIAL UPLOAD

Reviews 

Clarity: ★★★★★ Feasibility: ★★★★★ Availability: ★★★★★ 25/6/2025

Clarity: ★★☆☆☆ Feasibility: ★★★★★ Availability: ★☆☆☆☆ 25/6/2025



Add Review

Lessons Clarity Exam Feasibility Teacher Availability

★★★★☆ ★★★★★ ★★★★★

 Comment

new Review



Algoritmi e Strutture Dati

Professor Mario Rossi

Lessons Clarity ⓘ Exam Feasibility ⓘ Teacher Availability ⓘ

★★★★☆ ★★★★★ ★★★★★

Reviews 

Clarity: ★★★★★ Feasibility: ★★★★★ Availability: ★★★★★ 25/6/2025

Clarity: ★★☆☆☆ Feasibility: ★★★★★ Availability: ★☆☆☆☆ 25/6/2025

Clarity: ★★★★★ Feasibility: ★★★★★ Availability: ★★★★★ 25/6/2025

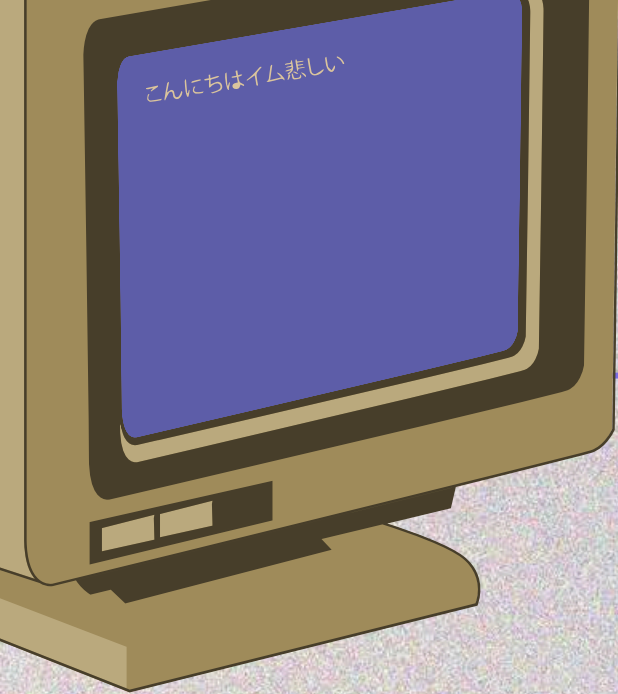


Challenge: *Lack of Confirmation Feedback*

- Users were uncertain if actions were completed
- **Most critical during:**
 - Material uploads
 - Review submissions

Solution: *Confirmation Notifications*

- Introduced explicit feedback mechanisms
 - Clear confirmation for material uploads
 - Clear confirmation for review submissions
- Ensures users know their actions are processed and recorded



CONTROLLED EXPERIMENT

Challenge: *Identifying the Best Review Reporting UI*

- **Goal:** Compare two interface options:
 - **Three-dots menu (:)** → reveals “Report”
 - **Direct “Report” button on each review**
- Need for objective performance data over design intuition

Solution: *Controlled Experiment Design*

- **Method:** Within-subjects experiment
 - Each participant tested both interfaces

Sample: 15 Sapienza University students (ages 19–25)

Demographics: Balanced gender; mix of Bachelor's and Master's students

Recruitment: From computer science, engineering, and library areas

Design: Split into two groups to avoid task familiarity bias

- **Group A (8):** Three-dots menu → Direct button
- **Group B (7):** Direct button → Three-dots menu

Independent Variable:

Interface Design (2 levels):

- **Interface 1:** Three-dots menu (:) with dropdown “Report”
- **Interface 2:** Direct “Report” button on each review

Dependent Variable:

Task Completion Time (in seconds) — from task start to successful report submission

Null Hypothesis (H_0):

No difference in performance — task completion times are equal for both interfaces.

Research Hypothesis (H_1):

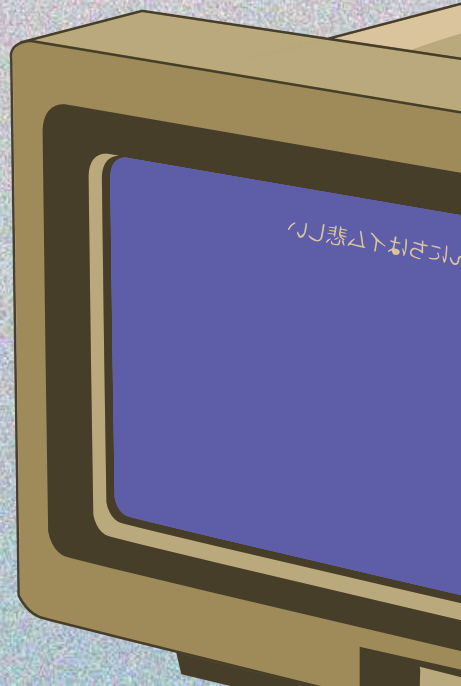
Direct “Report” button (Interface 2) leads to faster task completion than the three-dots menu (Interface 1), due to a simpler and more intuitive design.

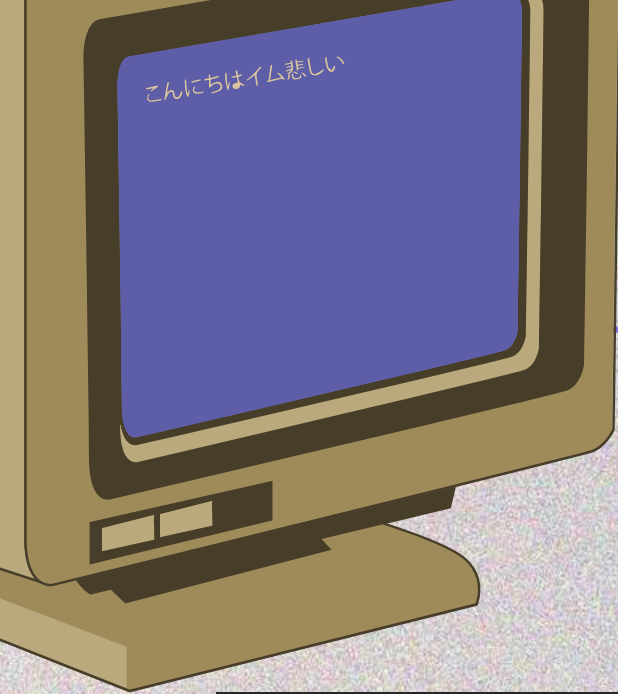
Experimental Procedure

- Group A (8): Interface 1 → 5-min break → Interface 2
- Group B (7): Interface 2 → 5-min break → Interface 1

Task Scenario:

“You find a review with inappropriate language and false course info. Report it to platform admins.”





CONTROLLED EXPERIMENT

ANOVA Single Factor Analysis

Descriptive Statistics:

- Interface 1 Mean: 15.360 seconds (SD = 1.15)
- Interface 2 Mean: 12.033 seconds (SD = 0.81)
- Mean Difference: 3.327 seconds (21.66% improvement)

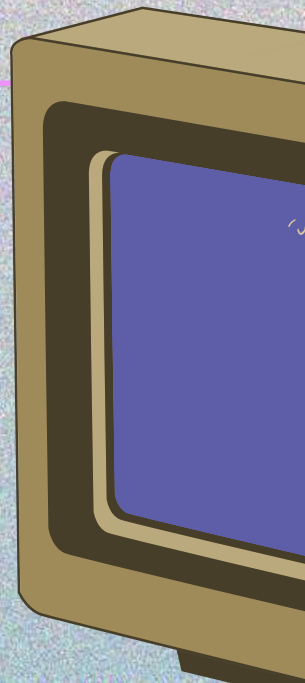
ANOVA Summary Table:

Participant	Interface 1 (seconds)	Interface 2 (seconds)
P1	15.2	12.4
P2	16.8	11.7
P3	13.5	13.2
P4	16.3	12.9
P5	15.1	10.8
P6	15.9	11.5
P7	13.7	12.6
P8	14.4	11.9
Participant	Interface 1 (seconds)	Interface 2 (seconds)
P9	17.2	13.1
P10	14.5	10.7
P11	16.9	12.3
P12	15.8	11.4
P13	15.6	12.8
P14	14.2	11.2
P15	15.3	12.0

Source	Sum of Squares	df	Mean Square	F	F-critical	p-value
Between Groups	83.02	1	83.02	83.91	4.196	< 0.001
Within Groups	27.70	28	0.99			
Total	110.72	29				

Note: F and p-values are only calculated for "Between Groups" as this tests our hypothesis. F-critical ($\alpha = 0.05$, $df_1 = 1$, $df_2 = 28$) = 4.196

Effect Size: $\eta^2 = 0.750$ (75% of variance explained by interface design)



CONTROLLED EXPERIMENT

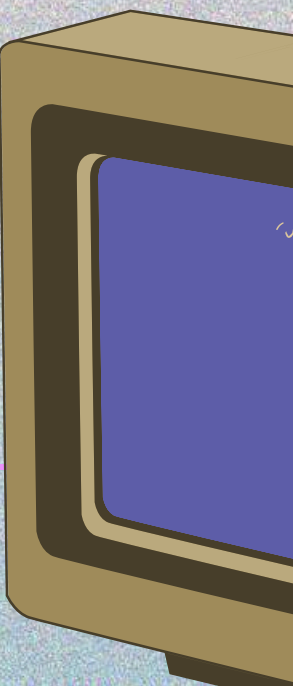
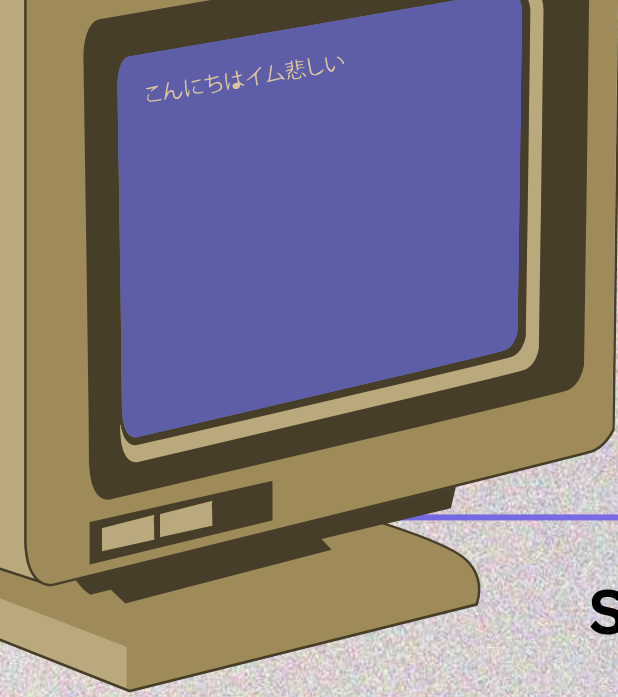
Results Interpretation

Statistical Significance: $F > F_{crit}$ ($83.91 > 4.196$)

- **Null Hypothesis:** REJECTED
- **Statistical Relevance:** Confirmed ($p < 0.001$, highly significant)
- **Effect Causation:** Significant result caused by Independent Variable (interface design)
- **Large Effect Size:** 75% of performance variance explained by interface choice

Practical Significance:

- Direct button interface resulted in 21.66% faster task completion
- All 15 participants performed faster with Interface 2
- Very large effect size indicates strong practical importance
- Consistent performance improvement across all participants



FINAL PRODUCT

Features of the final product:

- User Registration and Authentication
- Personalized Dashboard
- Personal Content Management
- Faculties Overview
- Advanced Course Discovery
- Course Overview
- Multi-Dimensional Course Reviews
- Advanced Review Filtering
- Review Reporting
- Study Material Repository



FUTURE WORK

- **AI-Powered Personalization:** ML-based course recommendations using academic history, peer similarity, and performance data
- **University System Integration:** APIs for real-time course updates and auto enrollment verification via official systems
- **Gamification Elements:** Reputation system & incentives to boost engagement and content quality



**SAPIENZA
ADVISOR**

FINAL PROTOTYPE

